Emmanuel Opoku-Adjei

Curriculum Vitae

Joint School of Nanoscience and Nanoengineering · University of North Carolina at Greensboro

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Google Scholar: scholar.google.com/citations?user=_1tc9A8AAAAJ · ResearchGate: researchgate.net/profile/Emmanuel-Opoku-Adjei

RESEARCH INTERESTS

Antimicrobial resistance at the interface between environmental reservoirs and patient infection; microbial adhesion and surface conditioning as determinants of colonisation; quantitative assay development for adhesion and resistance phenotypes; cross-kingdom fungal and bacterial interactions.

EDUCATION

Master of Science, Nanoscience Expected Spring 2027

Joint School of Nanoscience and Nanoengineering, University of North Carolina at Greensboro, NC, USA

- Cumulative GPA: 4.00 / 4.00 (in progress; degree expected Spring 2027)
- Thesis advisor: Professor Dennis LaJeunesse
- Thesis: *Surface and cell-side determinants of microbial adhesion: Candida secretome conditioning and sub-lethal disinfectant exposure on a common mineral substrate* (working title)

Bachelor of Science, Nursing — First Class Honours August 2022

Valley View University, Oyibi–Accra, Ghana

- Cumulative GPA: 3.77 / 4.00 (summa cum laude)
- Valedictorian — Overall Best Student in Academic Work, 2022 graduating class (university-wide)
- Best Graduating Student, Bachelor of Science in Nursing, 2022

RESEARCH EXPERIENCE

Graduate Research Assistant, Nanoscience August 2025 – present

Joint School of Nanoscience and Nanoengineering, UNC Greensboro · Advisor: Prof. Dennis LaJeunesse

Surface and cell-side determinants of microbial adhesion: Candida secretome conditioning and sub-lethal disinfectant exposure on a common mineral substrate

- Designed and built the Conditioned Surface Adhesion Assay (CSAA): conditioning-film deposition on muscovite mica microplatelets (17 to 25 micrometres), a five-arm control set including a no-bead direct-exposure reference, and two orthogonal readouts, OD600 and GFP fluorescence.
- Aim 1: Whether sub-threshold cleaning-agent exposure changes adhesion in five selected C. auris clinical isolates, by concentration series at fixed contact time.
- Aim 2: Whether Candida secretome conditioning changes E. coli adhesion, and whether the effect is surface-mediated or solution-mediated. Conditioning cleared the threshold on both readouts.
- Organisms: Aim 1, five C. auris clinical isolates. Aim 2, Candida albicans ATCC 90028 and three C. auris strains spanning adhesion phenotypes as secretome donors, with Escherichia coli pBTK501 (GFP) as the adhering organism, over seventeen formal runs across four strain blocks.
- Wrote the pre-registered analysis plan for Aim 2 and keep a documented deviations log. Exclusions and sensitivity checks follow rules set in advance.

Collaborative projects

- Beads Adhesion Precipitation (BAP) Assay — second author on the development and validation of a rapid method quantifying microbe-surface adhesion by OD600-based precipitation indexing.

Laboratory and analytical practice

- Operate and maintain a UV-Vis spectrophotometer for routine OD600 quantification and growth monitoring, fluorescence and optical microscopy, and a Synergy H1 multi-mode plate reader.

— Analyze and visualize multi-condition experimental data in Python (NumPy, Pandas, Matplotlib), GraphPad Prism and SPSS.

Research Assistant and Teaching Assistant (part-time), Department of Nursing September 2022 – June 2024
Valley View University, Accra, Ghana · Supervisor: Prof. Stella Appiah

— Contributed to qualitative and quantitative health-services research: study design, data collection, statistical analysis and manuscript preparation.

— Co-author on the resulting peer-reviewed publication, with responsibility for data reporting and technical writing.

— Served concurrently as Teaching Assistant, supporting undergraduate nursing instruction, laboratory sessions and student assessment across two academic years.

Undergraduate Research Project 2021 – 2022
Valley View University, Department of Nursing, Ghana · Supervisors: Prof. Stella Appiah and Ms. Dorothy Baffour Awuah

— Designed and executed a quantitative descriptive cross-sectional study of 212 respondents: sample size determination, sampling strategy, instrument design, validity and reliability testing, pretesting, and analysis in SPSS. This methodological grounding led directly to the research assistantship above.

PUBLICATIONS

Peer-reviewed, published

1. Osei, E. A., Addasah, F., Oware, J., Ani-Amponsah, M., Acquah, A., Amoah, E. O., Afoakwa-Acheampong, C., & **Opoku-Adjei, E.** (2025). Prioritizing care and addressing distinct needs of older adults with disabilities in Ghana amidst challenges. *Geriatric Nursing*, 65, 103524. doi:10.1016/j.gerinurse.2025.103524. **PMID: 40684472**.

Manuscripts submitted

— Karim, M. A., **Opoku-Adjei, E**., Mclaughlin, L., Adegoke, S. C., & LaJeunesse, D. Beads Adhesion Precipitation (BAP) Assay: a simple and rapid method to quantify microbe/surface adhesion. Submitted to Current Protocols.

Manuscripts in preparation

— **Opoku-Adjei, E.** & LaJeunesse, D. Surface and cell-side determinants of microbial adhesion: Candida secretome conditioning and sub-lethal disinfectant exposure on a common mineral substrate. First author; from thesis work.

— Impact of 0.1% hydrogen peroxide exposure on adhesion of clinical Candida auris isolates. Co-author.

Preprints on public servers

— None to date.

PRESENTATIONS

Oral presentations — first and presenting author

1. Opoku-Adjei, E. (2026, February 20). *Bridging clinical care and nanoscience: research progress*. Invited oral presentation, Graduate Research Retreat, Joint School of Nanoscience and Nanoengineering, University of North Carolina at Greensboro, Greensboro, NC.

— Selected by the Chair of the Nanoscience Department as the sole MS-level representative speaker for the department.

Conference abstracts submitted

— None to date.

CLINICAL EXPERIENCE

Nursing Officer — Orthopaedic and Post-Surgical Care July 2024 – July 2025
FOCOS Orthopaedic Hospital, Pantang–Accra, Ghana

— Delivered post-operative care to patients with implanted orthopaedic hardware, including prosthetic joints, fracture-fixation devices and spinal instrumentation.

— Monitored surgical sites and device-associated complications; recognised, documented and escalated adverse events under institutional protocol.

Registered General Nurse — Medical-Surgical January 2023 – January 2024
Shai Osudoku District Hospital, Ghana

— Provided medical-surgical nursing care under documented protocols, with responsibility for infection prevention and antimicrobial administration.

— Participated in quality-improvement initiatives linking evidence-based practice to patient-safety outcomes; mentored students and junior staff.

Student Clinical Rotations (seven placements) 2018 – 2021
Valley View University Department of Nursing, Ghana

HONOURS AND AWARDS

Valedictorian — Overall Best Student in Academic Work 2022
Valley View University · awarded to the highest-achieving student in the entire graduating class, across all disciplines.

Best Graduating Student, BSc Nursing 2022
Valley View University · awarded to the top graduate of the nursing cohort.

First Class Honours (summa cum laude), GPA 3.77 2022
Valley View University

CERTIFICATIONS AND PROFESSIONAL TRAINING

— CITI Program — Responsible Conduct of Research, Biomedical. Completed June 2026, valid to June 2029. Record ID 77725082, under University of North Carolina at Greensboro requirements.

— CITI Program — Good Laboratory Practice. In progress.

— IHI Open School Basic Certificate in Quality and Safety — Institute for Healthcare Improvement, August 2025 (17.75 hours: patient safety, improvement capability, human factors, responding to adverse events).

— Nursing and Midwifery Council of Ghana continuing professional development, January 2026 — 29 CPD credit points across three accredited courses (Liverpool School of Tropical Medicine; UNFPA Asia Pacific; Primary Care International).

— Continuing Professional Development — National Blood Service Ghana: Medico-Legal Considerations in Medical Decision Making for Blood Transfusion, May 2025.

— Quality Improvement Leadership Certificate (ACQUIRE) — Institute for Healthcare Improvement, 2025.

PROFESSIONAL LICENSURE

— Registered General Nurse — Nursing and Midwifery Council of Ghana. Registration No. RGN65022, valid to December 2028.

— NCLEX-RN — Authorization to Test issued; examination planned for January 2027. Arizona Board of Nursing licensure by examination in progress. Credential evaluation completed (Josef Silny & Associates).

TECHNICAL SKILLS

— Instrumentation. UV-Vis spectrophotometry, routine OD600 quantification and growth curves; Synergy H1 multi-mode microplate reader; fluorescence and optical microscopy. Scanning electron microscopy training scheduled.

— **Laboratory.** Quantitative microbial adhesion assays; fungal and bacterial culture (Candida spp., Escherichia coli); GFP-reporter fluorescence assays; surface conditioning protocols; aseptic technique; assay design with defined control architecture.

— **Computational and analytical.** Python (NumPy, Pandas, Matplotlib, scikit-learn); GraphPad Prism; SPSS; Excel; Git; experimental design, pre-registration and statistical comparison of multi-condition data.

— **Clinical.** Electronic health record and laboratory information systems; adverse-event recognition and escalation; SOP-driven documentation in regulated environments; infection prevention practice.

REFERENCES

Professor Dennis LaJeunesse
Thesis Advisor, Joint School of Nanoscience and Nanoengineering, University of North Carolina at Greensboro
Additional references available on request.